

Does a Short Questionnaire Improve AI Skin-Concern Classification, and for Whom?

Working title: Questionnaire-Augmented Skin-Concern Classification under Missing Answers: A Skin-Tone-Stratified Study

Status: DRAFT v0.1, for discussion with the supervising professor. Items marked [VERIFY] must be checked before they are used in a paper.

1. Research question

In plain words: When a user answers a few simple questions (how long the problem has lasted, symptoms, skin type), does the AI get better at guessing their skin problem from a photo? What happens when the user skips some questions? And does the AI work equally well for light and dark skin?

Formal sub-questions

- **RQ1 (benefit)**. Does adding self-reported questionnaire data to a photo-based classifier improve classification of common skin concerns, compared with the photo alone?
- **RQ2 (missing answers)**. How does performance change when some or all questionnaire answers are missing, and can training the model for missing answers (e.g., modality dropout) keep it reliable?
- **RQ3 (fairness)**. Is the benefit of the questionnaire, and the loss caused by missing answers, the same across skin tones (Fitzpatrick and Monk groups)?
- **RQ4 (optional, secondary)**. When the model says "uncertain, see a professional", is it uncertain equally often and equally correctly across skin tones?

2. Why this matters

Most published fusion work (image plus patient information) uses skin-cancer datasets such as HAM10000 and ISIC. Everyday concerns photographed with ordinary phones are less studied. Real users also skip questions, and public skin datasets have uneven skin-tone coverage, so an average accuracy number can hide unfair behavior. A careful study of these three issues together would help anyone building consumer skin-analysis tools.

3. Related work and the gap (to be completed after reading)

Area	What exists	Gap this study targets
Image + metadata fusion	Many studies on lesion/cancer datasets; some on SCIN (e.g., multi-label transformer with metadata) [VERIFY details]	Missing/noisy answers and skin-tone breakdowns on everyday-concern data
Missing-modality robustness	Modality dropout and gated fusion shown on ISIC 2019	Not found on SCIN with skin-tone-stratified results [VERIFY with systematic search]
Skin-tone fairness	Fitzpatrick17k, DDI, SCIN used for fairness studies; foundation models evaluated across tones	Effect of <i>questionnaire data</i> on fairness is not clearly studied [VERIFY]
Abstention / selective prediction	Well studied generally; examples in lesion classification	Per-skin-tone coverage and error for consumer skin concerns [VERIFY]
Foundation models	PanDerm, DermLIP strong baselines	Must be included as baselines

Gap statement (draft): Existing work shows that adding patient metadata can help skin-image classifiers, but it mostly studies cancer-type lesions, assumes complete metadata, and reports averages. We study everyday skin concerns, deliberately remove questionnaire answers, and report results separately for each skin tone.

4. Data

- **Primary dataset: SCIN (Google)**. 10,000+ images from consented US contributors, with self-reported symptoms, texture, duration, demographics, dermatologist differential diagnoses, and estimated Fitzpatrick (eFST) and Monk (eMST) skin tones. About half of contributions include demographics and about 80% include condition information, so real missingness exists. [VERIFY license terms and current download]
- **Image-only and skin-tone evaluation:** DDI and Fitzpatrick17k have no questionnaire-style fields, so they can support image-only baselines and skin-tone analysis but not the fusion experiments.
- **Possible second dataset for fusion:** PAD-UFES-20 (smartphone images with clinical metadata), but it is lesion/cancer focused, so it would be a transfer check, not a replica of the main setting. [VERIFY suitability]
- **Label design (to do first):** Map SCIN diagnoses to a small, well-supported class set (for example acne, eczema/dermatitis, redness-type conditions, other) plus an "uncertain" option. **Count images per class before committing.** SCIN labels are dermatologists' retrospective differentials with confidence scores, not biopsy-proven truth. Define clearly how a single label is derived (e.g., top-weighted diagnosis above a confidence threshold).
- **Splitting:** Split by contributor/case, not by image, because one case can have up to three photos. Otherwise the same person appears in training and test and results look better than they are.

5. Method

Questionnaire features. Use fields that match the app questionnaire: skin type, duration, symptoms (itching, burning, flaking...), texture, body area, age group, optional self-reported skin type. Encode categories and add an explicit "missing" indicator.

Models compared

1. Image-only baselines: (a) EfficientNet or MobileNetV3 fine-tuned (as in the thesis plan), (b) a dermatology foundation model (PanDerm or DermLIP) with a simple classifier on frozen features, (c) optionally a general vision model (e.g., DINOv3) as a control.
2. Questionnaire-only baseline (tabular model) to show how much the answers alone explain.
3. Fusion models: simple concatenation (late fusion), then one stronger method (gated or cross-attention fusion).
4. Missing-answer-aware fusion: the same fusion model trained with random modality/answer dropout.

Missing-answer experiment. At test time, remove a controlled share of answers (0%, 25%, 50%, 75%, 100%), randomly and also by field (e.g., drop symptoms only). Compare models trained with and without dropout. Also report results on SCIN's naturally missing cases.

Skin-tone analysis. Group by eFST and eMST (e.g., I-II, III-IV, V-VI). Report each group separately and the gap between best and worst group. Treat groups with very few images carefully and state sample sizes.

Metrics

- Balanced accuracy and macro-F1 (the data is imbalanced, so plain accuracy is misleading)
- Per-class and per-skin-tone results, and worst-group performance
- Calibration (ECE) and risk-coverage curves for the "uncertain" output (RQ4)
- Repeat with at least 5 random seeds; report mean, spread, and bootstrap confidence intervals; use paired tests when comparing models

6. Experiment plan

#	Experiment	Answers
E1	Image-only vs questionnaire-only vs fusion on SCIN	RQ1
E2	Fusion with 0-100% answers removed; with and without dropout training	RQ2
E3	Results by Fitzpatrick/Monk group for E1 and E2	RQ3
E4	Confidence/abstention analysis per skin-tone group	RQ4
E5	Foundation-model vs EfficientNet/MobileNet backbones	Strength of conclusions
E6	Transfer check on a second dataset (if feasible)	Generality

7. Expected contributions

1. A controlled comparison of image-only, questionnaire-only, and fused models on everyday-concern data.
2. Evidence of how fast performance drops when answers are missing, and whether dropout training helps.
3. A skin-tone-stratified analysis of who benefits from the questionnaire and who does not.
4. Public code and exact data splits for reproducibility.

8. Risks and limitations

- **Label noise:** SCIN labels are retrospective differentials. Results measure agreement with dermatologist labels, not true diagnosis.
- **Small groups:** Darker skin tones may have few images in some classes, giving wide confidence intervals.
- **Questionnaire mismatch:** SCIN's self-reported fields are not identical to the app's questionnaire; document the mapping.
- **Novelty risk:** Similar work may exist or appear soon. A systematic search is needed before committing.
- **Not a medical tool:** The study is informational, with no diagnostic claims.

9. Next steps

1. Read the three closest papers in full (marked in section 10).
2. Run a systematic literature search (Google Scholar, Semantic Scholar, arXiv).
3. Download SCIN, count images per class and per skin-tone group, and decide the class set.
4. Run E1 first (quick baseline) to see whether the questionnaire helps at all.
5. Review this draft with the professor and agree the target journal and scope.

10. References and links

I only read the abstract or a snippet of several of these. Open each one and confirm details before citing.

Read first (closest to this study)

1. **SCIN dataset: GitHub and data description**. <https://github.com/google-research-datasets/scin>
2. **SCIN overview (Google Research blog)**. <https://research.google/blog/scin-a-new-resource-for-representative-dermatology-images/>
3. **Multi-label classification analysis with modified C-Tran on SCIN dataset** (uses SCIN with metadata inputs). https://www.researchgate.net/publication/395180569_Multi-label_classification_analysis_with_modified_C-Tran_on_SCIN_dataset
4. **Benchmarking Off-the-Shelf Multimodal AI Models Against Dermatologists on Patient-Captured Skin Images** (SCIN, with and without patient metadata). <https://arxiv.org/html/2609.24190>
5. **Disease Burden over Skin Tone: Decomposing the Dermatology-AI Generalization Gap** (DDI and SCIN, foundation models). <https://arxiv.org/html/2609.02111>

Baseline models

6. **PanDerm: A multimodal vision foundation model for clinical dermatology** (Nature Medicine). <https://www.nature.com/articles/s41591-025-03747-y>
7. **DermIM / DermLIP** (dermatology vision-language models). <https://arxiv.org/html/2503.14911v2>

Multimodal fusion and missing data

8. **Comparative analysis of multimodal architectures for skin lesion detection using clinical and image data** (HAM10000, fusion strategies). <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12424468/>
9. **Pre- and Post-Gated Attention-based Multimodal Fusion for Skin Lesion Classification** (modality dropout, ISIC 2019). <https://www.researchsquare.com/article/rs-7455733/v1>
10. **Attention-guided multimodal skin disease classification using discounted Dempster-Shafer fusion and EfficientNet**. <https://www.sciencedirect.com/science/article/pii/S2352914825000887>

Fairness and skin tone

11. **Fitzpatrick 17k dataset and paper** (Groh et al., CVPR 2021). <https://github.com/mattgroh/fitzpatrick17k>
12. **Diverse Dermatology Images (DDI)** (Daneshjou et al., Science Advances 2022, "Disparities in dermatology AI performance on a diverse, curated clinical image set"). <https://ddi-dataset.github.io/>
13. **Reducing skin tone bias in dermatology AI via sketch-guided multimodal fusion** (Scientific Reports). <https://www.nature.com/articles/s41598-026-59316-x>

Uncertainty and abstention

14. **An Empirical Analysis of Calibration and Selective Prediction in Multimodal Clinical Condition Classification** (ICU data, useful for methods and metrics). <https://arxiv.org/html/2603.02719>
15. **Skin Lesion Classification in Low-Resource Settings Using Lightweight CNNs with Uncertainty Estimation** (EfficientNet-B0, MC Dropout, Grad-CAM). <https://www.mdpi.com/2673-7426/6/4/53>

Papers you mentioned (adjacent, cite as context)

16. **SkinCAP: A Multi-modal Dermatology Dataset Annotated with Rich Medical Captions**. <https://arxiv.org/abs/2405.18004>
17. **Deep learning-based skin care product recommendation: cosmetic ingredient analysis and facial skin conditions** (J Cosmet Dermatol). <https://onlinelibrary.wiley.com/doi/full/10.1111/jocd.16218>
18. **Demographically oriented facial attribute recognition for skincare and grooming** (ISGD / AKRTI) (The Visual Computer). <https://link.springer.com/article/10.1007/s00371-026-04613-9>

Search terms for the systematic literature search

- "SCIN" AND "metadata" AND ("missing" OR "incomplete")
- "skin tone" AND "selective prediction" AND dermatology
- "self-reported symptoms" AND "skin" AND "multimodal" AND fairness
- "modality dropout" AND dermatology
- "consumer" OR "smartphone" AND "skin condition" AND "Fitzpatrick" AND "fusion"